



Power BI

Business Intelligence

Business Intelligence (BI) refers to the technologies, strategies, and processes used by organizations to collect, analyze, and transform data into actionable insights. These insights help businesses make informed decisions, improve efficiency, and gain a competitive edge.

What is Power BI

Power BI is a Microsoft-powered business intelligence tool that helps transform raw data into interactive dashboards and actionable insights. It allow users to connect to various data sources, clean and shape data and visualize it using charts, graphs and reports all with minimal coding.

Power BI workflow

- Bring in data from different places like Excel files, SQL databases, CSVs, JSON files and even websites.
- Clean and fix your data.
- Create visuals like bar charts, line graphs, pie charts and dashboards to help you see patterns and trends.
- Analyze your data using filters and slicers, so you can focus on specific details like sales by region or product performance over time.
- Share your dashboards with your team or clients so everyone stays informed and can explore the data on their own.
- Set up automatic updates so your reports always show the latest information without you having to do anything.

Parts of Power BI

Power BI is not just a single tool it is a collection of multiple tools

- Power BI: Desktop
- Power BI: Service (SaaS)
- Power BI: Mobile Apps

What Is Power Bi Report Vs Dashboard

Report and a dashboard serve different purposes even though people often mix them up.

A **Report** is a detailed, multi-page analytical view created in Power BI Desktop. It is built from a single dataset and allows users to interact deeply with the data using slicers, filters, drill-through, and page navigation. Reports are mainly used for analysis and exploration, where users want to understand trends, compare values, and go into details.

A **Dashboard** is a single-page summary created only in the Power BI Service (online). It is made by pinning visuals from one or multiple reports and even from different datasets. Dashboards are designed for high-level monitoring and decision-making, showing KPIs, key metrics, and important insights at a glance. They are less interactive than reports but are ideal for executives and stakeholders who want a quick overview rather than deep analysis.

KPI

A KPI (Key Performance Indicator) is a measurable value that shows how effectively a person, team, or organization is achieving an important goal. It helps track performance and tells whether you are moving in the right direction or not.

ETL

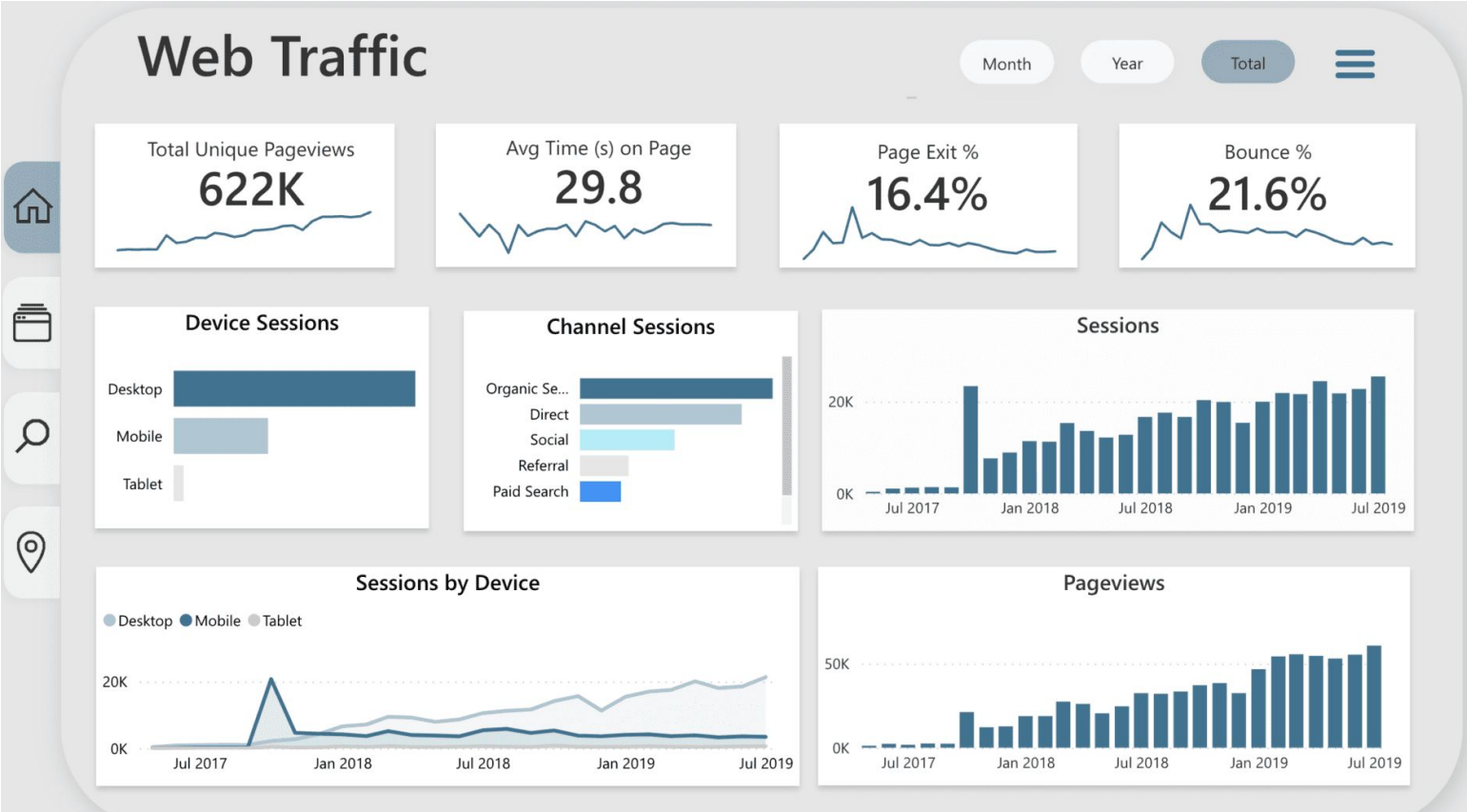
ETL process stands for Extract, Transform, Load. It is a data pipeline used to collect data from different sources, clean and process it, and then store it in a place where it can be analyzed easily, such as a data warehouse or database.

Extract means pulling raw data from multiple sources like databases, Excel files, APIs, logs, or cloud applications. The data at this stage can be in different formats and may contain errors or missing values.

Transform means cleaning and converting the extracted data into a usable format. This includes removing duplicates, handling missing values, changing data types, standardizing formats (like dates or currencies), applying business rules, and sometimes creating new calculated fields. This step ensures the data is accurate and consistent.

Load means storing the transformed data into a target system such as a data warehouse, data lake, or analytics database. This data is then used for reporting, dashboards, and analysis in tools like Power BI, Tableau, or Excel.

Report



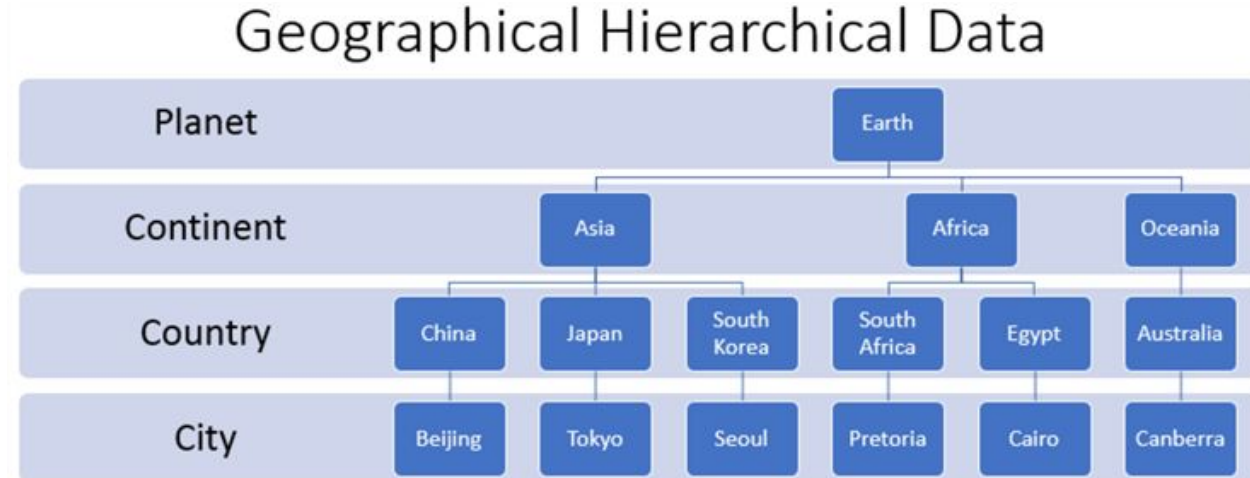
Power BI Installation

- Microsoft App Store
- Website - Search power BI download and download from official website

Hierarchy

Hierarchy means arranging related fields in a parent-child order so that you can drill down from a broader level to a more detailed level in a visual.

- Hierarchy helps because it makes reports cleaner, more interactive, and easier to analyze.
- In business reports, common hierarchies are Geography (Country → State → City), Product (Category → Sub-category → Product), and Time (Year → Month → Date).



What Is Table Relationship

Table relationship is the connection between two tables that tells Power BI how data in one table is related to data in another table. This relationship is usually created using a common column, such as a Customer ID, Product ID, or Date, and it allows Power BI to combine data from multiple tables so you can analyze them together in reports and visuals.

Cardinality

cardinality describes the nature of the relationship between two tables based on how many times a value in one table can appear compared to the other table. In simple words, it explains how many rows in one table are related to rows in another table.

- **one-to-many (1:*)**, where a single value in one table appears multiple times in another table.
- **many-to-one (*:1)**, which is basically the same relationship but viewed from the opposite direction.
- **one-to-one (1:1)**, where each value in one table matches exactly one value in another table.
- **many-to-many (:)**, where values in both tables can repeat.

What Is Data Model

A model (also called a data model) is the structured way in which your data is organized inside Power BI so that it can be analyzed correctly and efficiently. It is not just the raw data you load, but the complete setup that defines how different tables work together.

A Power BI model consists of tables, the relationships between those tables, columns, calculated columns, measures, and hierarchies.

What Is A Fact And Dimension Table

A fact table stores the business events or transactions that you want to analyze. It usually contains numeric values that can be aggregated, such as sales amount, quantity, discount, profit, or cost. A fact table also includes foreign keys like Customer ID, Product ID, or Date ID, which link it to dimension tables. Fact tables are typically very large and grow continuously because they record each transaction or event.

A dimension table stores descriptive information that gives context to the facts. It contains attributes such as customer name, product name, category, region, or date details like year, month, and quarter. Dimension tables are usually smaller, change less frequently, and have a unique key that connects to the fact table. These tables are used for filtering, grouping, and labeling data in reports.

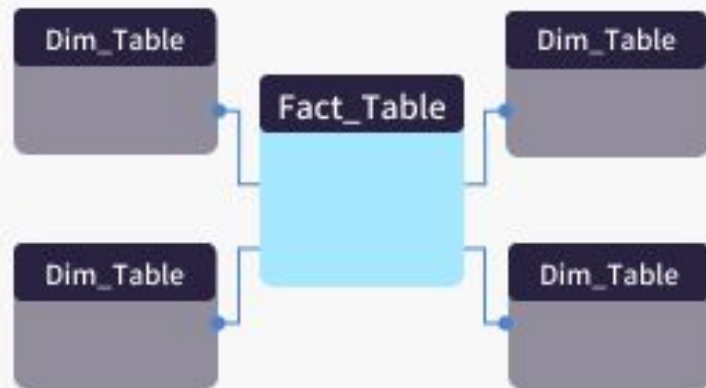
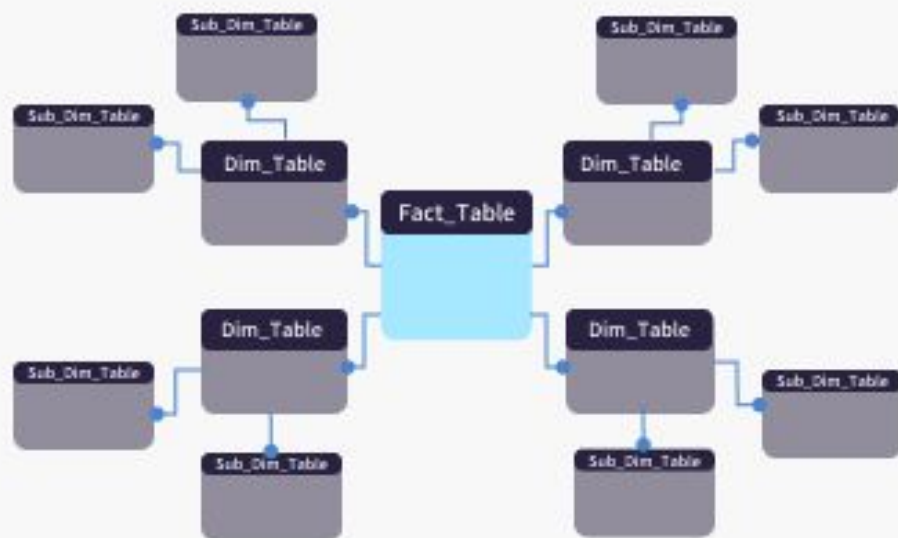
Star Schema

star schema model, which is the most used and most efficient model in Power BI. In this model, you have one central fact table (for example, Sales) that contains numeric values like sales, quantity, and profit, and several surrounding dimension tables (such as Customers, Products, Date, Region). Each dimension table has a one-to-many relationship with the fact table. This structure is simple, performs very well, and works perfectly with DAX and Power BI's storage engine.

Snowflakes Schema

The snowflake schema model is similar to the star schema, but dimension tables are further broken into sub-dimension tables. For example, a Product table may be split into Category and Subcategory tables. While this reduces data redundancy, it increases the number of relationships and can slightly impact performance and complexity, so it is less preferred than star schema in Power BI.

Snowflake Schema vs Star Schema



What Is DAX?

Data Analysis Expressions (DAX) DAX (Data Analysis Expressions) is a powerful programming language designed for data analysis and calculations. It uses formulas and functions to manipulate data and derive meaningful insights. Widely used in major Microsoft products like Power BI, Excel Power Pivot, and SQL Server Analysis Services (SSAS), DAX is use to create dynamic and complex data models and It optimized performance allows it to handle large datasets efficiently

Types Of Calculations In DAX.

- Measures
- Calculated Column
- Calculated Table

Measure

A **Measure** is a calculation that is performed on the fly when you use it in a visual. It does not store values in the table. It calculates results based on the current filter context.

For example, Total Sales = SUM(Sales[Amount]).

If you change filters (like year or region), the result changes automatically.

Measures are mainly used for aggregations like SUM, AVERAGE, COUNT, Profit %, etc. Measures are dynamic.

Calculated Column

A Calculated Column is created inside a table and calculates a value for each row. The result is stored in the model. It is calculated during data refresh.

For example, $\text{Profit} = \text{Sales}[\text{Sales Amount}] - \text{Sales}[\text{Cost}]$.

Each row gets its own value.

Calculated columns are static until the next refresh. They are mostly used for row-level logic.

Calculated Table

A Calculated Table creates a completely new table using DAX. For example, you can create a summary table using `SUMMARIZE()` or filter a table using `FILTER()`. It is also calculated during data refresh and stored in the model.

Types of Measures in Power BI

Implicit Measures - Implicit measures are automatically created by Power BI when you drag a numeric field into a visualization. These measures are simple and convenient for quick aggregations but offer limited flexibility.

Example: If you drag the Sales Amount column from the Sales table into a chart, Power BI will automatically create an implicit measure that sums the sales amounts.

Explicit Measures - Explicit measures are user-defined calculations created using DAX (Data Analysis Expressions). These measures provide more control and flexibility, allowing you to define complex calculations tailored to your specific needs.

Example: Total Sales Measure $\text{Total Sales} = \text{SUM}(\text{Sales}[\text{Sales Amount}])$

Calculated columns

- Basic calculation among columns
- For text columns
 - UPPER
 - LOWER
 - LEFT
 - RIGHT
 - MID
 - CONCAT
- ROUND

Calculated columns

- IF
 - AND
 - OR
- SWITCH
- RELATED - *vlookup of DAX*

Measures

- SUM
- MIN
- MAX
- AVERAGE
- COUNT
 - COUNTBLANK
 - COUNTROWS
- SUMX
- MINX
- MAXX
- AVERAGEX
- CALCULATE - *Filter+Aggregation*

Calculated Table

- CALENDAR
 - YEAR
 - MONTH
 - DAY
 - WEEKDAY
 - FORMAT
- DATEDIFF
- TODAY
- NOW
- SUMMARIZE